```
# Task1
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_salary():
    e1 = int(emp1.get())
    e2 = int(emp2.get())
    e3 = int(emp3.get())

    data = {
        "Employee":["A","B","C"],
        "Salary":[e1,e2,e3]
    }

    df = pd.DataFrame(data)

    avg = np.mean(df["Salary"])
    high = np.max(df["Salary"])
    low = np.min(df["Salary"])

    result.config(text=f"Average Salary: {avg}\nHighest Salary: {high}\nLowest Salary: {low}")

    plt.bar(df["Employee"],df["Salary"])
    plt.title("Employee Salary Distribution")
    plt.xlabel("Employees")
    plt.ylabel("Salary")
    plt.show()

root = tk.Tk()
root.title("Employee Salary Analyzer")

tk.Label(root,text="Employee A Salary").pack()
emp1 = tk.Entry(root)
```

```
emp1.pack()
tk.Label(root,text="Employee B Salary").pack()
emp2 = tk.Entry(root)
emp2.pack()

tk.Label(root,text="Employee C Salary").pack()
emp3 = tk.Entry(root)
emp3.pack()

tk.Button(root,text="Analyze Salary",command=analyze_salary).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

 # Task2

```
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_units():
    jan = int(january.get())
    feb = int(february.get())
    mar = int(march.get())

    data = {
        "Month":["Jan","Feb","Mar"],
        "Units":[jan,feb,mar]
    }

    df = pd.DataFrame(data)

    total = np.sum(df["Units"])
```

```
    avg = np.mean(df["Units"])

    result.config(text=f"Total Units: {total}\nAverage Units: {avg}")

    plt.plot(df["Month"],df["Units"],marker="o")
    plt.title("Electricity Usage")
    plt.xlabel("Month")
    plt.ylabel("Units")
    plt.show()

root = tk.Tk()
root.title("Electricity Bill Analyzer")

tk.Label(root,text="January Units").pack()
january = tk.Entry(root)
january.pack()

tk.Label(root,text="February Units").pack()
february = tk.Entry(root)
february.pack()

tk.Label(root,text="March Units").pack()
march = tk.Entry(root)
march.pack()

tk.Button(root,text="Analyze Usage",command=analyze_units).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

```
# Task3
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_steps():
    d1 = int(day1.get())
    d2 = int(day2.get())
    d3 = int(day3.get())

    data = {
        "Day":["Mon","Tue","Wed"],
        "Steps":[d1,d2,d3]
    }

    df = pd.DataFrame(data)

    avg = np.mean(df["Steps"])

    result.config(text=f"Average Steps: {avg}")

    plt.plot(df["Day"],df["Steps"],marker="o")
    plt.title("Fitness Progress")
    plt.xlabel("Days")
    plt.ylabel("Steps")
    plt.show()

root = tk.Tk()
root.title("Fitness Activity Tracker")

tk.Label(root,text="Monday Steps").pack()
day1 = tk.Entry(root)
day1.pack()
```

```
tk.Label(root,text="Tuesday Steps").pack()
day2 = tk.Entry(root)
day2.pack()

tk.Label(root,text="Wednesday Steps").pack()
day3 = tk.Entry(root)
day3.pack()

tk.Button(root,text="Analyze Activity",command=analyze_steps).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

 # Task4

```
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_attendance():
    total = int(total_classes.get())
    attended = int(attended_classes.get())

    percentage = (attended/total)*100

    result.config(text=f"Attendance Percentage: {percentage:.2f}%")

    labels = ["Attended","Missed"]
    values = [attended,total-attended]

    plt.pie(values,labels=labels,autopct='%1.1f%%')
    plt.title("Attendance Distribution")
    plt.show()
```

```
root = tk.Tk()
root.title("Attendance Analyzer")

tk.Label(root,text="Total Classes").pack()
total_classes = tk.Entry(root)
total_classes.pack()

tk.Label(root,text="Classes Attended").pack()
attended_classes = tk.Entry(root)
attended_classes.pack()

tk.Button(root,text="Analyze Attendance",command=analyze_attendance).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

 # Task5

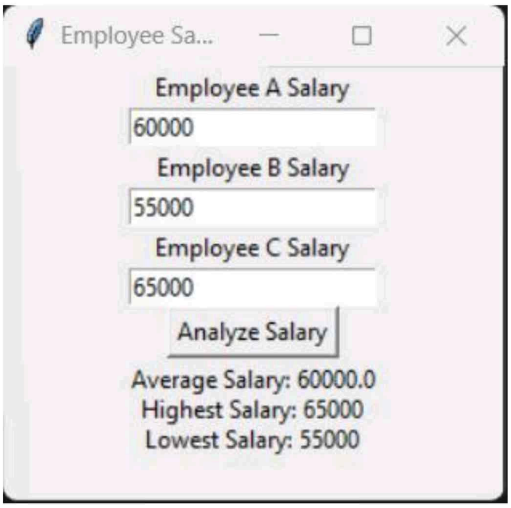
```
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_score():
    p1 = int(player1.get())
    p2 = int(player2.get())
    p3 = int(player3.get())

    data = {
        "Player":["A","B","C"],
        "Score":[p1,p2,p3]
    }
```

```
    df = pd.DataFrame(data)

    avg = np.mean(df["Score"])

    result.config(text=f"Average Score: {avg}")

    plt.bar(df["Player"],df["Score"])
    plt.title("Cricket Score Comparison")
    plt.xlabel("Players")
    plt.ylabel("Score")
    plt.show()

root = tk.Tk()
root.title("Cricket Score Analyzer")

tk.Label(root,text="Player A Score").pack()
player1 = tk.Entry(root)
player1.pack()

tk.Label(root,text="Player B Score").pack()
player2 = tk.Entry(root)
player2.pack()

tk.Label(root,text="Player C Score").pack()
player3 = tk.Entry(root)
player3.pack()

tk.Button(root,text="Analyze Scores",command=analyze_score).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

```
# Task6
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_water():
    d1 = float(day1.get())
    d2 = float(day2.get())
    d3 = float(day3.get())

    data = {
        "Day":["Mon","Tue","Wed"],
        "Water":[d1,d2,d3]
    }

    df = pd.DataFrame(data)

    avg = np.mean(df["Water"])

    result.config(text=f"Average Water Intake: {avg} Liters")

    plt.plot(df["Day"],df["Water"],marker="o")
    plt.title("Water Intake Tracker")
    plt.xlabel("Days")
    plt.ylabel("Liters")
    plt.show()

root = tk.Tk()
root.title("Water Intake Tracker")

tk.Label(root,text="Monday Intake (L)").pack()
day1 = tk.Entry(root)
day1.pack()
```

```
tk.Label(root,text="Tuesday Intake (L)").pack()
day2 = tk.Entry(root)
day2.pack()

tk.Label(root,text="Wednesday Intake (L)").pack()
day3 = tk.Entry(root)
day3.pack()

tk.Button(root,text="Analyze Water Intake",command=analyze_water).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

Outputs:



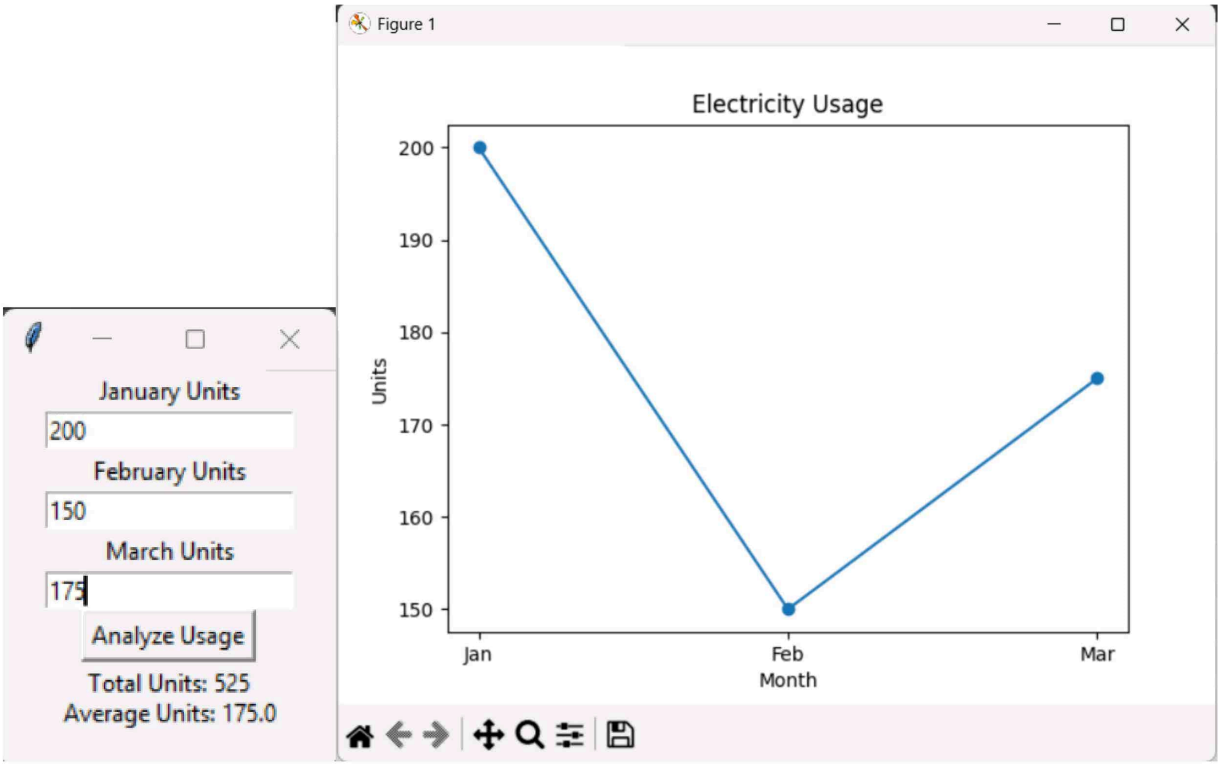


Figure 1
Electricity Usage
Units
200
190
180
170
160
150
Jan
Feb
Mar
Month
January Units
200
February Units
150
March Units
175
Analyze Usage
Total Units: 525
Average Units: 175.0

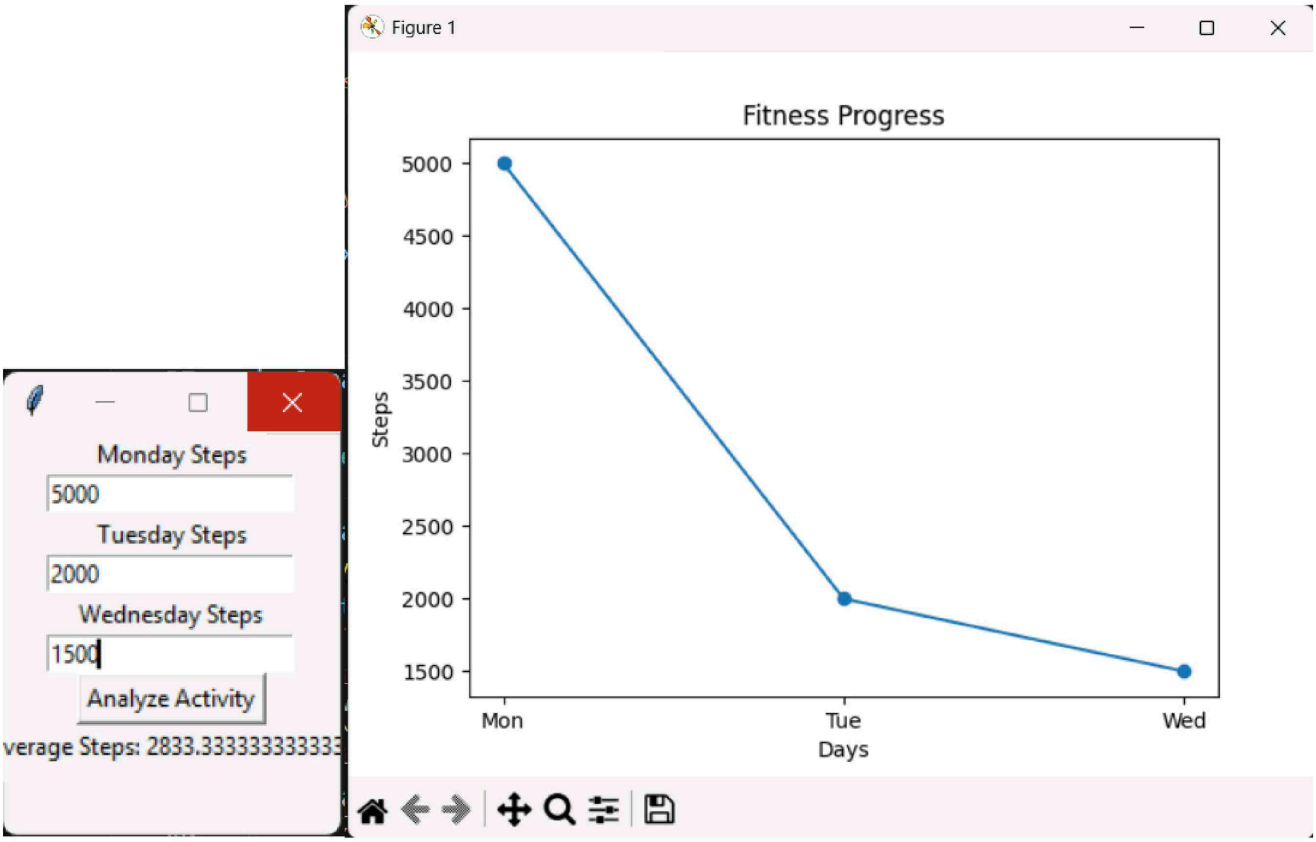
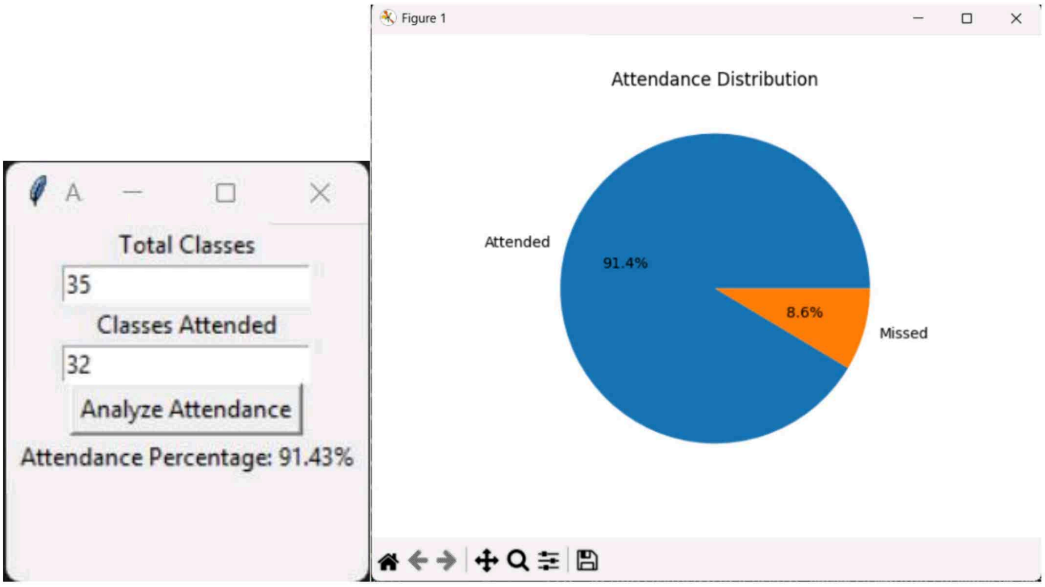
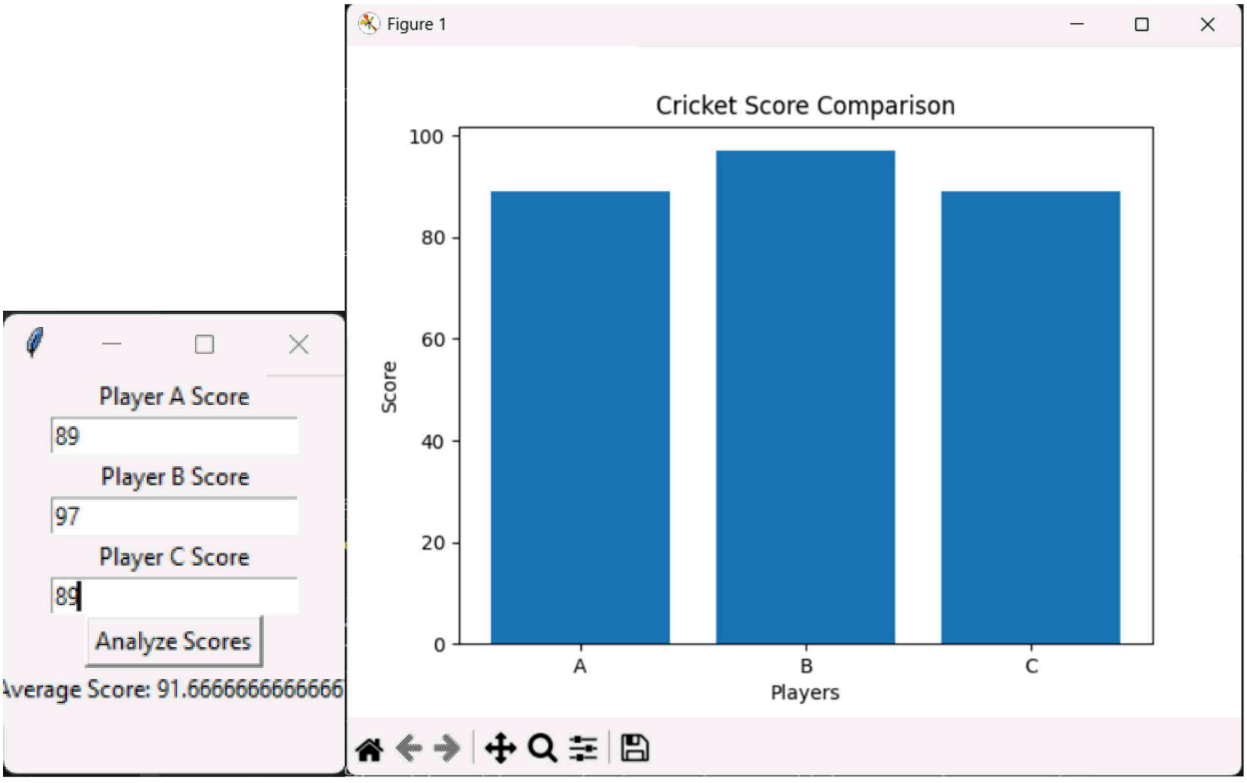
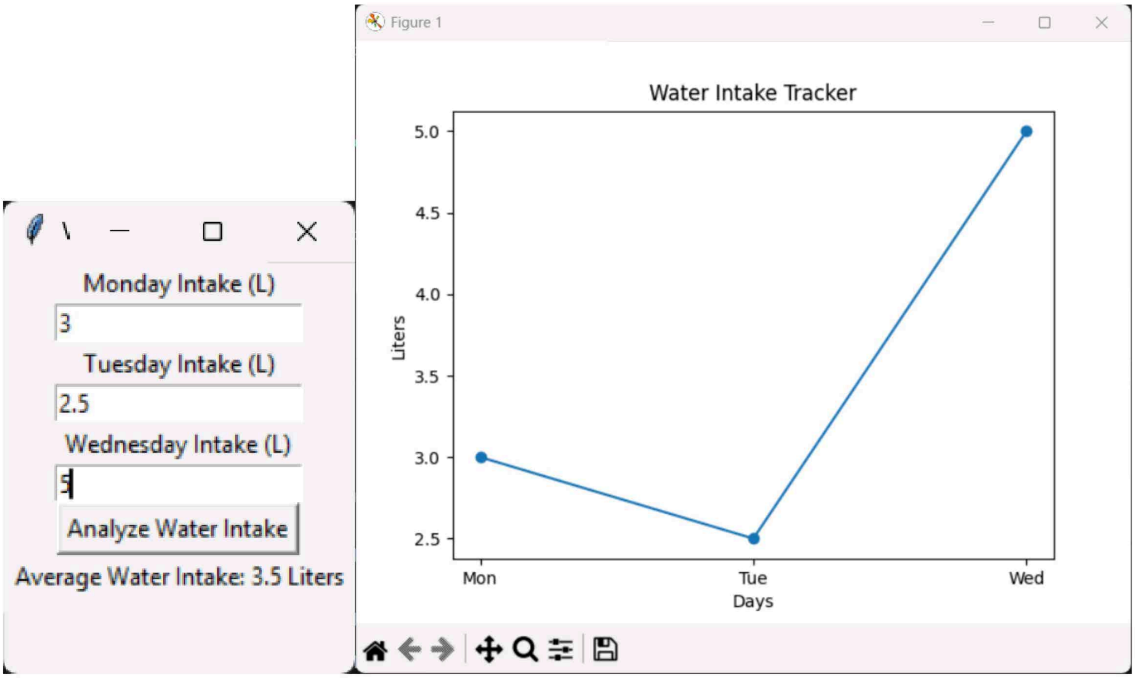


Figure 1
Fitness Progress
Steps
5000
4500
4000
3500
3000
2500
2000
1500
Mon
Tue
Wed
Days
Monday Steps
5000
Tuesday Steps
2000
Wednesday Steps
1500
Analyze Activity
verage Steps: 2833.333333333333


Total Classes
35
Classes Attended
32
Analyze Attendance
Attendance Percentage: 91.43%
Figure 1
Attendance Distribution
Attended
91.4%
8.6%
Missed


Player A Score
89
Player B Score
97
Player C Score
89
Analyze Scores
Average Score: 91.6666666666666
Figure 1
Cricket Score Comparison
100
80
60
40
20
0
Score
A
B
C
Players


Monday Intake (L)
3
Tuesday Intake (L)
2.5
Wednesday Intake (L)
5
Analyze Water Intake
Average Water Intake: 3.5 Liters
Figure 1
Water Intake Tracker
5.0
4.5
4.0
3.5
3.0
2.5
Liters
Mon
Tue
Wed
Days